CIE AS PHYSICS (9702) • QUESTION PAPER

3.2 Non-Uniform Motion

Syllabus Section: 3. Dynamics **Paper Type:** Multiple Choice (Paper 1) **Total:** 15 Questions (15 Marks)

Question 1 Easy [1 mark]

An egg is dropped from the top of a three-storey building. It falls through air until it reaches the ground

What remains constant throughout the fall?

A acceleration of the egg

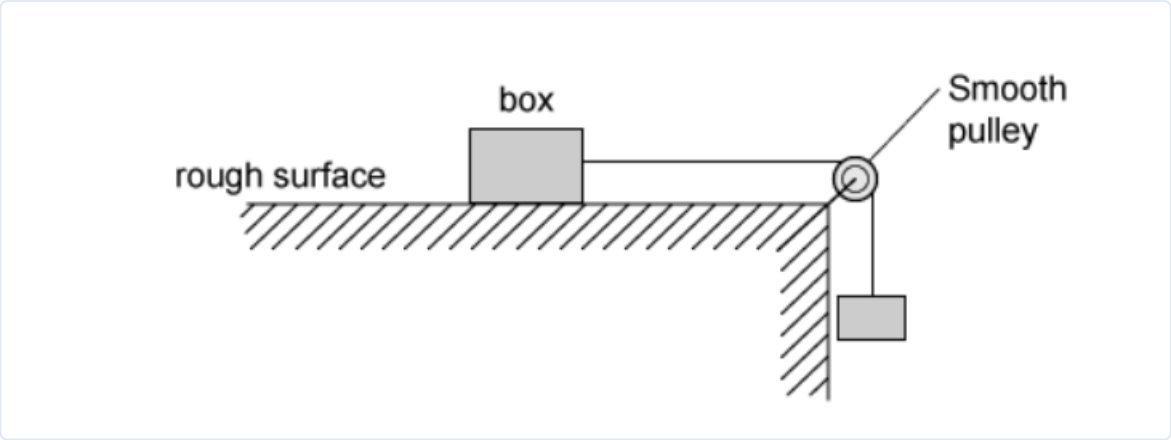
B weight of the egg

C velocity of the egg

D air resistance on the egg

Question 2 Easy [1 mark]

A box rests on a horizontal, rough surface. A string attached to the box passes over a smooth pulley and supports a mass at its other end.



When the box is released, what forces are acting on it?

A weight only

B friction and weight

C friction, weight and reaction

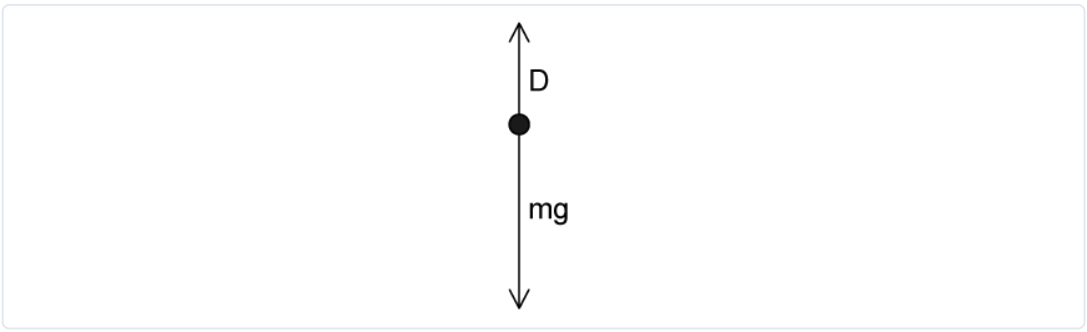
D tension, friction, weight and reaction

Question 3 Easy [1 mark]

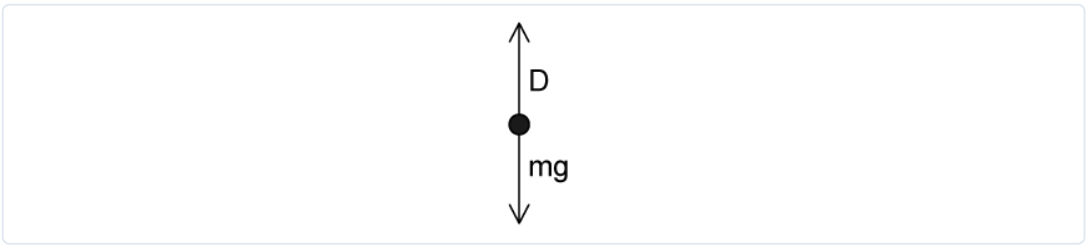
An object of weight *mg* falls through the air and experiences a drag force *D*.

Which of the following free-body diagrams shows the object when it has reached terminal velocity?

A



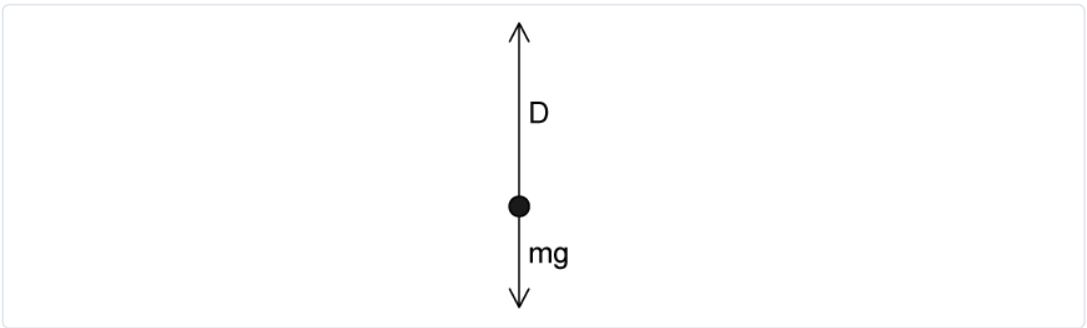
B



C



D



Question 4 Easy [1 mark]

A skydiver has a weight of 800 N, and experiences a drag force of 450 N.

What is the magnitude and direction of the resultant force of the skydiver?

	Magnitude / N	Direction
A	1250	Downwards
B	350	Upwards
C	1250	Upwards
D	350	Downwards

A

B

C

D

Question 5 Easy [1 mark]

A lorry is driving along a horizontal road and experiences a resistive force opposing the forward motion.

Assuming that air resistance is negligible, what is the name of the resistive force?

A Upthrust

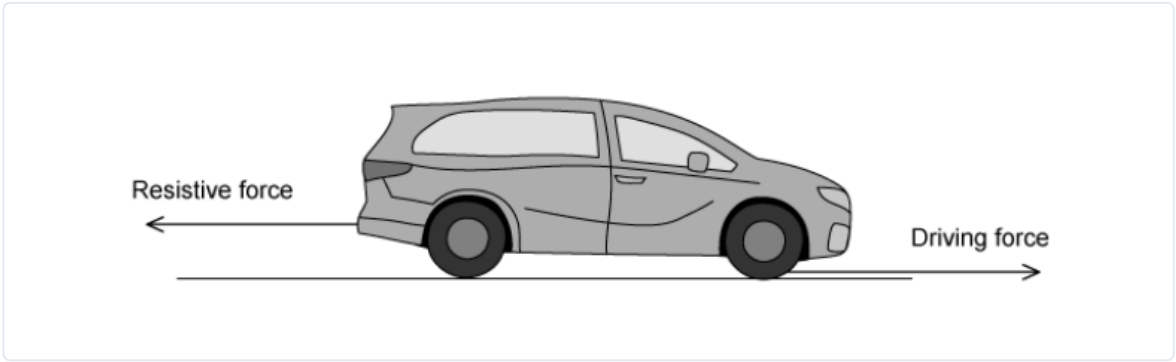
B Weight

C Friction

D Normal contact force

Question 6 Medium [1 mark]

A car has a horizontal driving force of 2.0 kN acting on it and a resistive force a quarter of this size. It has a forward horizontal acceleration of 2.0 m s^{-2}



What is the mass of the car?

A 500 kg

B 750 kg

C 1250 kg

D 2000 kg

Question 7 Medium [1 mark]

A stone of mass *m* is dropped from a tall building. There is significant air resistance. The acceleration of free fall is *g*.

When the stone reaches its terminal velocity, which row is correct?

	Magnitude of the acceleration of the stone	Magnitude of the force of gravity on the stone	Magnitude of the force of air resistance on the stone
A	zero	*mg*	*mg*
B	*g*	*mg*	*mg*
C	*g*	zero	zero
D	zero	zero	zero

A

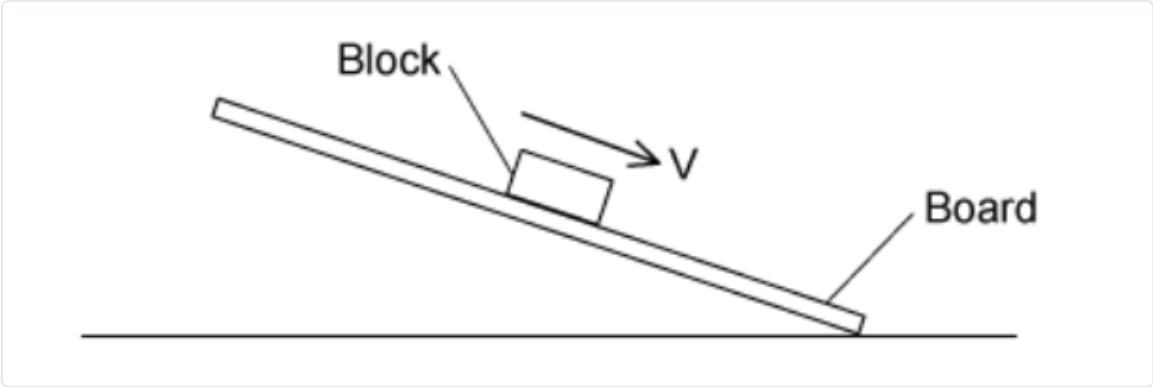
B

C

D

Question 8 Medium [1 mark]

A wooden block rests on a rough board. The end of the board is then raised until the block slides down the plane of the board at constant velocity *v*



Which row describes the forces acting on the block when sliding with constant velocity?

	Frictional force on block	Resultant force on block
A	down the plane	zero
B	down the plane	down the plane
C	up the plane	zero
D	up the plane	down the plane

A

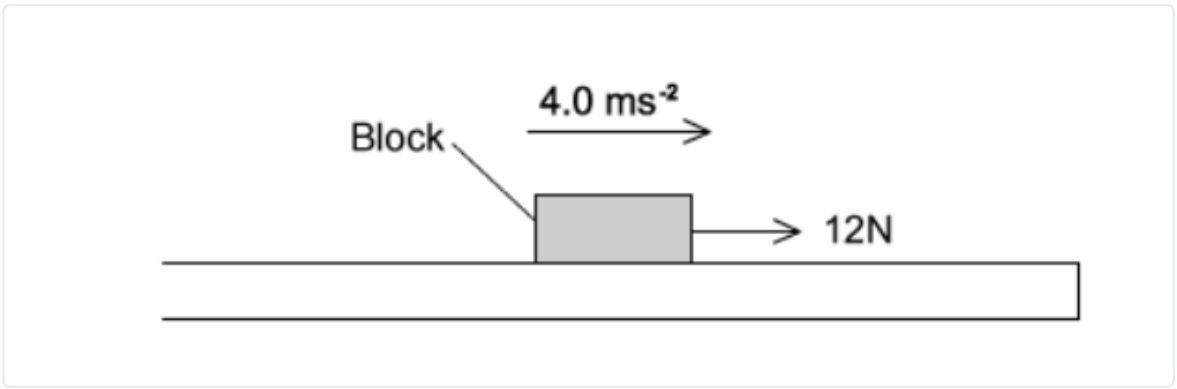
B

C

D

Question 9 Medium [1 mark]

A block of mass 1.20 kg is on a rough horizontal surface. A force of 12 N is applied to the block and it accelerates at 4.0 m s^{-2}



What is the magnitude of the frictional force acting on the block?

A 2.4 N

B 3.6 N

C 7.2 N

D 16.8 N

Question 10 Medium [1 mark]

A boat travels forward in a straight line. The driving force of the boat is less than the water resistance acting on the boat.

Which statement is true?

A The boat is travelling at constant velocity

B The resultant force acts in the same direction as the motion

C The boat is decelerating

D The resultant force is equal to the driving force of the boat

Question 11 Hard [1 mark]

The three forces acting on a hot air balloon as it moves vertically are its weight, the force due to air resistance and upthrust.

The hot air balloon descends vertically at constant speed. The force of air resistance on the balloon is F.

What weight of material must be released from the balloon so that it ascends vertically at the same constant speed?

A $\frac{1}{2}F$

B F

C $2F$

D $4F$

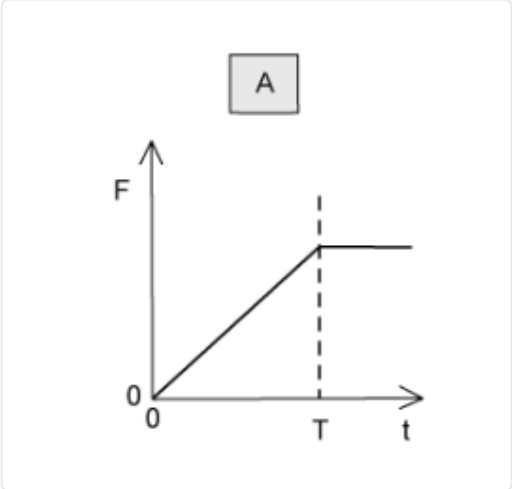
Question 12 Hard [1 mark]

A skydiver falls vertically from a stationary balloon.

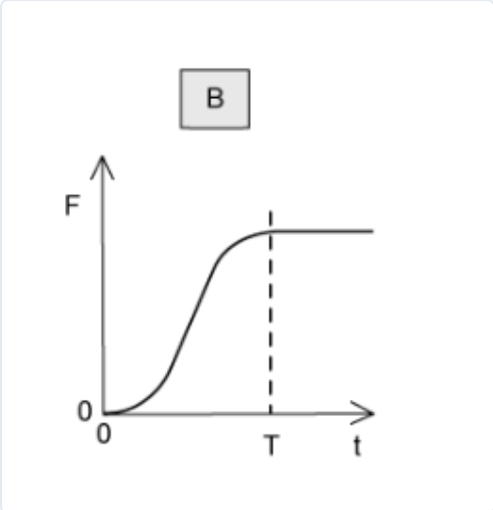
She leaves the balloon at time $t = 0$. At time $t = T$, she reaches terminal velocity. Beyond the time shown in the graphs, she opens her parachute

Which graph shows the variation with time t of the force F due to air resistance?

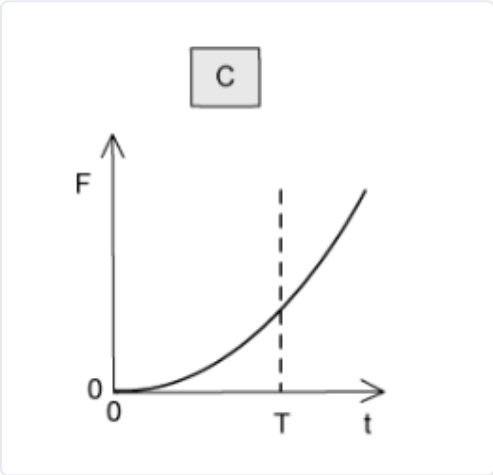
A



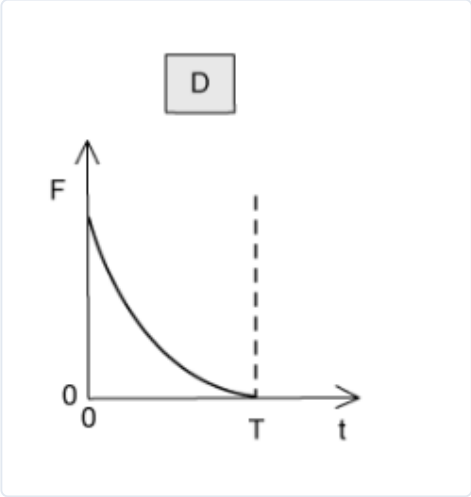
B



C



D



Question 13 Hard [1 mark]

Two skydivers, A and B, experience drag forces D_1 and D_2 respectively. Skydiver A has mass m and reaches terminal velocity v_1. Skydiver B has mass $2m$ and reaches terminal velocity v_2.

Given that $D \propto v^2$, what is the value of $\frac{v_1}{v_2}$?

A $\frac{1}{2}$

B $\frac{1}{4}$

C 2

D $\frac{1}{\sqrt{2}}$

Question 14 Hard [1 mark]

A car of mass m starts from rest and accelerates uniformly, reaching a velocity of v over distance s. The car experiences a frictional force of $\frac{mg}{2}$.

What is the driving force of the car?

A $\frac{m}{2}\frac{v^2}{s} + g$

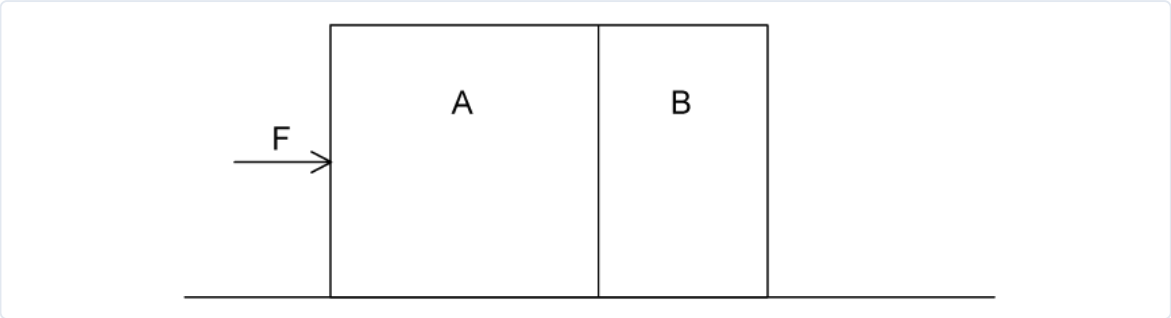
B $m\frac{v^2}{s} + \frac{W}{2}$

C $\frac{m}{2}\frac{v^2}{s} - g$

D $m\frac{v^2}{s} - g$

Question 15 Hard [1 mark]

The diagram shows two blocks, A and B, that have masses of $3m$ and $2m$ respectively. A force F is applied to Block A, and both blocks accelerate along a horizontal surface. Block B experiences a drag force of $\frac{F}{5}$.



What is the resultant force of Block A on Block B?

A $\frac{1}{5}F$

B $\frac{4}{25}F$

C $\frac{8}{25}F$

D $\frac{2}{5}F$